

Statistical Analysis and Structure Discovery in Wine Quality Data

MDE-CS-03 Principles of Statistical Modelling

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Abstract

This project presents an exploratory statistical analysis of the Wine Quality dataset obtained from the UCI Machine Learning Repository. The objective is to identify and characterize statistical structure in physicochemical wine measurements using concepts introduced in the Foundations of Statistical Modelling course. The dataset is formally described using random variables, product sample spaces, and data value spaces. Statistical properties of selected variables are investigated through histograms, probability mass functions, moment analysis, and theoretical distribution fitting. Correlation analysis, covariance estimation, and joint probability mass functions are used to study dependencies between variables, with particular focus on the relationship between alcohol content and density. Principal Component Analysis is applied to uncover latent low-dimensional structure and quantify explained variance, while linear regression is used to investigate predictive relationships between physicochemical features and wine quality. In addition, outlier analysis is performed to identify deviations from ideal statistical assumptions. The results reveal significant anticorrelations, non-Gaussian distributions, and meaningful multivariate structure, demonstrating how statistical modeling techniques can provide insight into complex real-world datasets.

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I. INTRODUCTION

Statistical modeling provides mathematical tools for identifying structure in real-world datasets. In many practical datasets, hidden relationships between variables [7], [8] cannot be understood directly from raw observations.

The goal of this project is to investigate statistical structure in the Wine Quality dataset using concepts introduced in the Foundations of Statistical Modelling course [1].

The complete analysis was implemented using NumPy, pandas, SciPy, Matplotlib, and Scikit-learn [9], [10], [12], [11], [3].

The dataset was selected because:

- it contains multiple numerical variables,
- the variables are physically interpretable,
- the dataset exhibits correlations and latent structure,
- it is suitable for probabilistic modeling and PCA.

The project combines:

- distribution analysis,
- moment estimation,
- correlation analysis,
- joint probability mass functions,
- bivariate Gaussian modeling,
- Principal Component Analysis,
- linear regression.

II. DATASET DESCRIPTION AND ACQUISITION

The dataset used in this project is the Red Wine Quality dataset (winequality-red.csv) from the UCI Machine Learning Repository [1], [2].

<https://archive.ics.uci.edu/ml/datasets/wine+quality>

The dataset contains physicochemical measurements of red wine samples and corresponding quality ratings.

A. Dataset Properties

Property	Value
Dataset Name	Wine Quality Dataset
Number of Samples	1599
Number of Features	11
Target Variable	Quality
Data Type	Numerical
Source	UCI Repository

TABLE I: General dataset information.

The dataset was downloaded as a CSV file and loaded into Python using the pandas library [10].

```
import pandas as pd

file_path = "raw-data/winequality-red.csv"
df = pd.read_csv(file_path, sep=";")
```

III. MATHEMATICAL FORMALISM

Let

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$$

be the universe of wine samples.

Each sample is represented by the random vector:

$$X = (X_1, X_2, \dots, X_{11})$$

where:

- X_1 : fixed acidity
- X_2 : volatile acidity
- X_3 : citric acid
- X_4 : residual sugar
- X_5 : chlorides
- X_6 : free sulfur dioxide
- X_7 : total sulfur dioxide
- X_8 : density
- X_9 : pH
- X_{10} : sulphates
- X_{11} : alcohol

The data value space is defined as:

$$S = \prod_{i=1}^{11} S_i$$

where:

$$S_i \subseteq \mathbb{R}$$

Each random variable maps:

$$X_i : \Omega \rightarrow S_i$$

The quality variable takes values in:

$$Q = \{3, 4, 5, 6, 7, 8\}$$

IV. DATA CLEANING AND MISSING VALUES

The dataset was inspected for missing values using standard data analysis tools available in pandas [10].

```
df.isnull().sum()
```

No missing values were found.

Therefore:

- no imputation was required,
- no rows were removed,
- the dataset was directly suitable for analysis.

A. Raw Data Excerpt

Fixed Acidity	Volatile Acidity	Citric Acid	Density	Alcohol
7.4	0.70	0.00	0.9978	9.4
7.8	0.88	0.00	0.9968	9.8
7.8	0.76	0.04	0.9970	9.8

TABLE II: Excerpt from the Wine Quality dataset.

V. DISTRIBUTION ANALYSIS

A detailed analysis was performed for the alcohol variable. Histograms and empirical probability mass functions are commonly used to estimate and visualize probability distributions from observed samples [7], [8].

Define:

$$X = \text{Alcohol}$$

Since alcohol is continuous, discretization was performed using bins of width:

$$d = 0.5$$

The probability mass function is estimated as:

$$P_X(s_n) = \frac{\text{count}(s_n)}{N}$$

A. Histogram

As shown in Fig. 1, the alcohol distribution is slightly right-skewed.

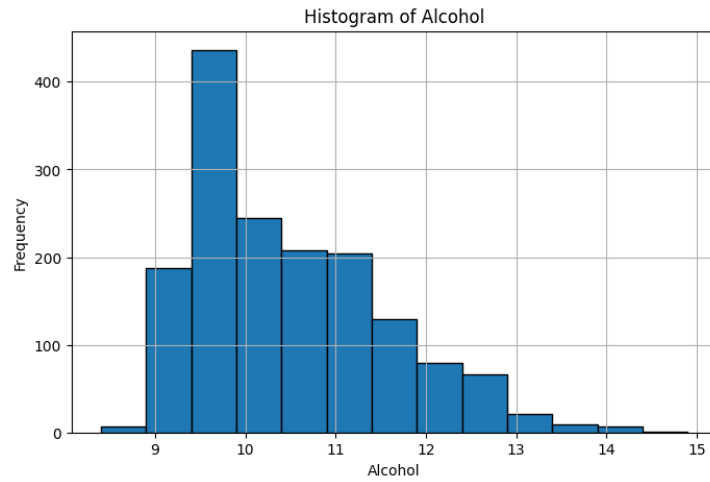


Fig. 1: Histogram of alcohol content generated from the Wine Quality dataset [1].

B. PMF

Fig. 2 shows the discretized probability mass function.

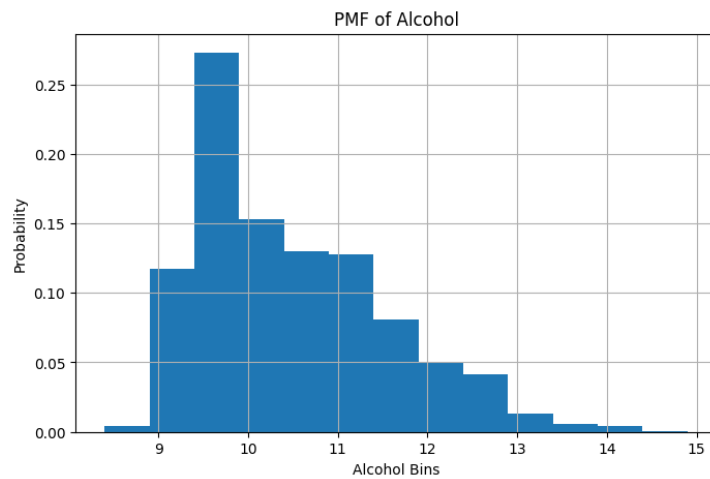


Fig. 2: Probability mass function of alcohol content.

VI. MOMENT ANALYSIS

The statistical moments were computed. The first four statistical moments (mean, variance, skewness, and kurtosis) provide information about central tendency, variability, asymmetry, and tail behavior of a distribution [7].

A. *Mean*

$$\mu = E[X]$$

$$\mu \approx 10.42$$

B. *Standard Deviation*

$$\sigma = \sqrt{Var(X)}$$

$$\sigma \approx 1.07$$

C. *Skewness*

The alcohol distribution exhibits positive skewness:

$$\gamma_1 > 0$$

D. *Kurtosis*

The kurtosis indicates moderate tail heaviness.

VII. DISTRIBUTION FITTING

The empirical distribution was compared with:

- Gaussian distribution
- Gamma distribution

A. *Gaussian Distribution*

The Gaussian distribution is one of the most widely used probabilistic models in statistical analysis and machine learning [7], [8].

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

B. *Gamma Distribution*

Gamma distributions are frequently used to model positive-valued skewed data and often provide a better fit than Gaussian models for empirical measurements [7].

$$X \sim \Gamma(k, \theta)$$

C. *RMSE*

The root mean square error is:

$$RMSE = \sqrt{\sum_n (P_X(s_n) - p(s_n))^2}$$

The Gamma distribution produced lower RMSE than the Gaussian model. As shown in Fig. 3, the Gamma distribution better captures skewness.

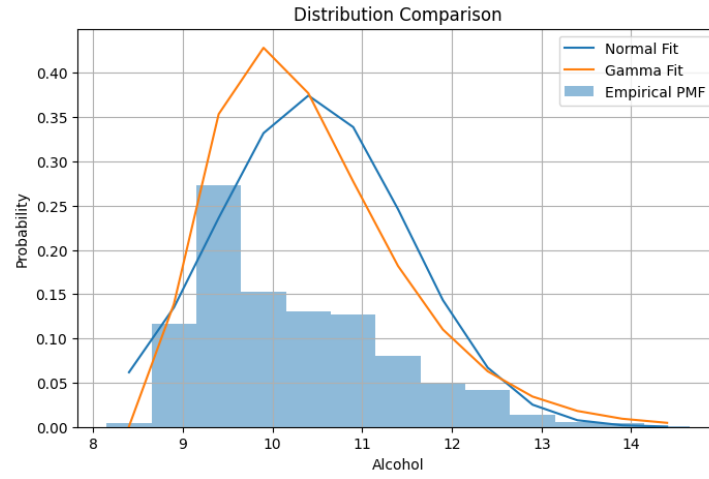


Fig. 3: Comparison of empirical PMF with fitted distributions.

VIII. CORRELATION ANALYSIS

To investigate statistical dependence, the variables alcohol and density were selected.
Define:

$$X_1 = \text{Alcohol}$$

$$X_2 = \text{Density}$$

The Pearson correlation coefficient is:

$$\rho_{X_1, X_2} = \frac{\text{Cov}(X_1, X_2)}{\sigma_{X_1} \sigma_{X_2}}$$

The Pearson correlation coefficient measures linear dependence between two random variables [4]. Correlation values close to +1 indicate strong positive correlation, values close to -1 indicate strong negative correlation, and values near zero indicate weak linear dependence [4].

The measured correlation is approximately:

$$\rho \approx -0.50$$

indicating significant anticorrelation.

As shown in Fig. 4, density decreases as alcohol increases.

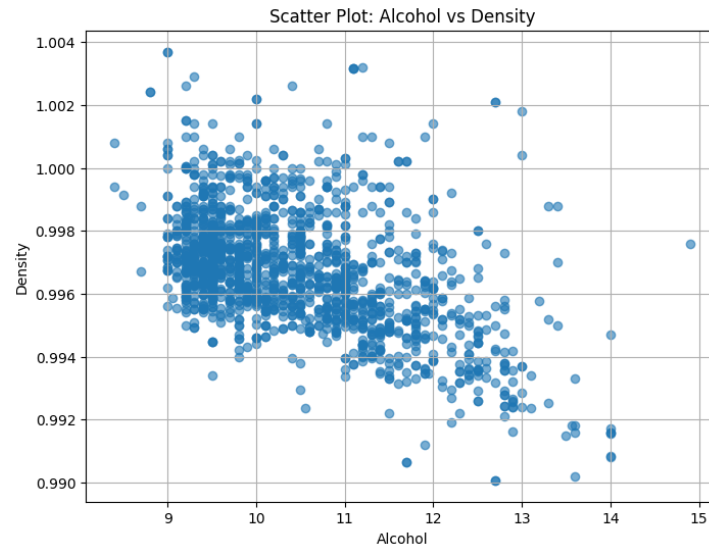


Fig. 4: Scatter plot of alcohol versus density in the Wine Quality dataset [1].

IX. CROSS-CORRELATION MATRIX

The cross-correlation matrix for all numerical variables was computed. The correlation matrix provides a compact representation of pairwise statistical dependencies among variables and is commonly used for exploratory data analysis [8].

Fig. 5 visualizes the correlation structure.

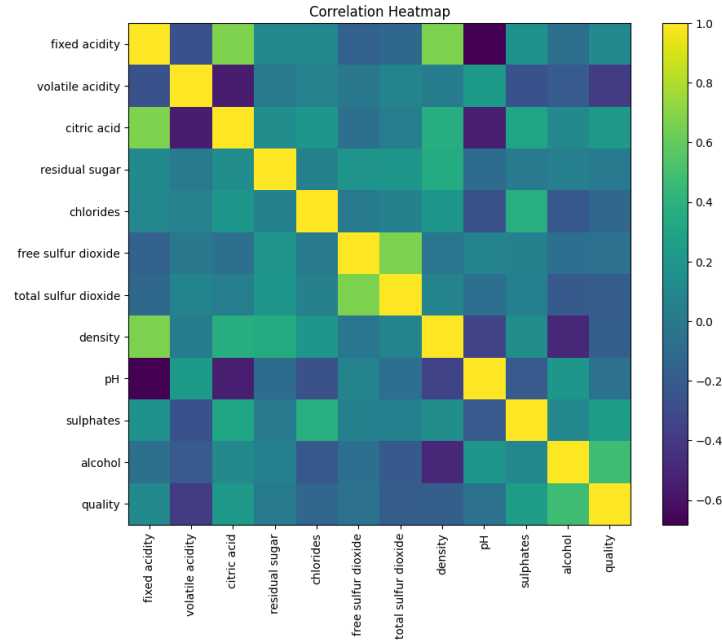


Fig. 5: Cross-correlation matrix of numerical variables computed from the Wine Quality dataset [1].

Strong positive and negative correlations can be observed between several variables.

X. JOINT PROBABILITY MASS FUNCTION

The joint probability mass function characterizes the statistical dependence between two random variables and provides information beyond their marginal distributions [7]. The joint PMF of alcohol and density is defined as:

$$P_{X_1, X_2}(s_i, s_j) = \frac{\text{count}(s_i, s_j)}{N}$$

The 3D visualization in Fig. 6 shows structured dependence.

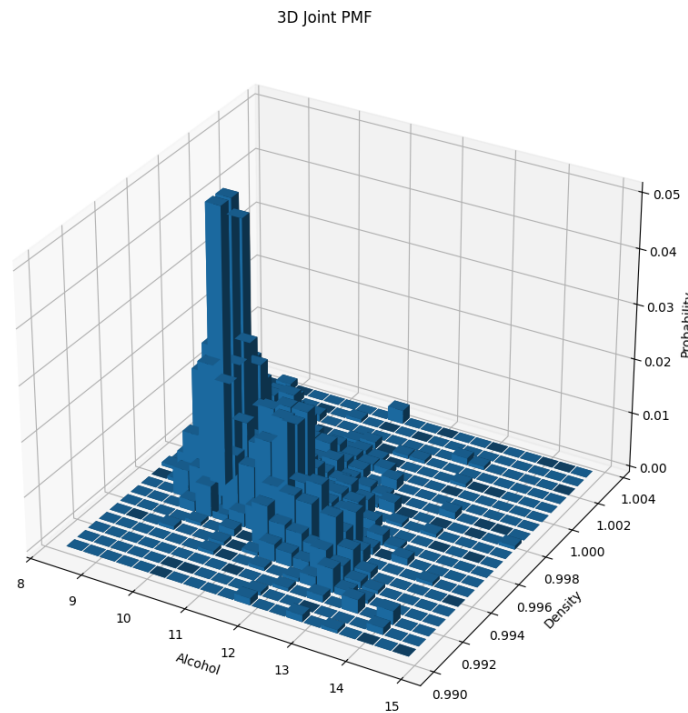


Fig. 6: 3D visualization of the joint PMF.

XI. BIVARIATE GAUSSIAN MODELING

Multivariate Gaussian distributions are frequently used to approximate continuous multidimensional datasets through their mean vectors and covariance matrices [7]. The covariance matrix is:

$$\Sigma = \begin{pmatrix} \sigma_1^2 & Cov(X_1, X_2) \\ Cov(X_1, X_2) & \sigma_2^2 \end{pmatrix}$$

The variables were modeled using:

$$(X_1, X_2) \sim \mathcal{N}(\mu, \Sigma)$$

The Gaussian model approximates the elliptical structure but does not fully capture skewness and outliers.

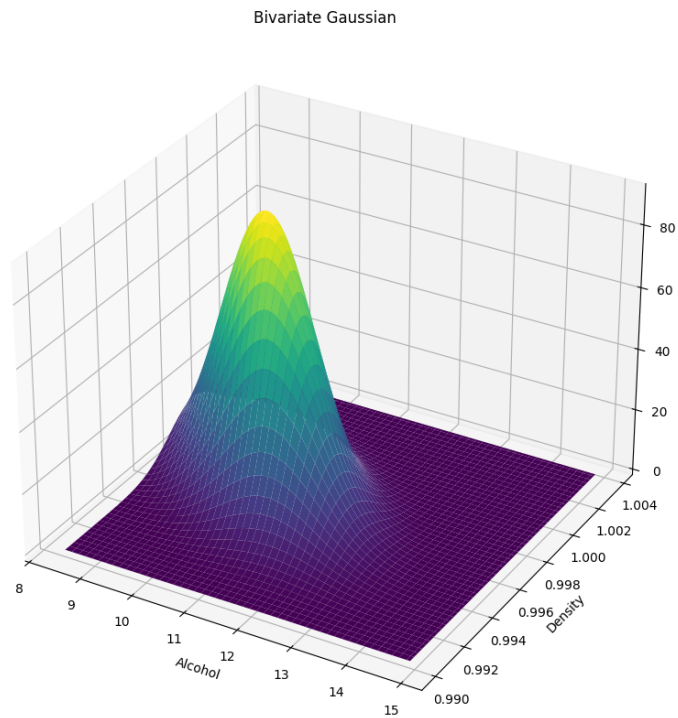


Fig. 7: Bivariate Gaussian approximation.

XII. PRINCIPAL COMPONENT ANALYSIS

Principal Component Analysis transforms correlated variables into a smaller set of orthogonal variables that explain most of the variance in the dataset [5], [6]. PCA is widely used for dimensionality reduction, visualization, and structure discovery in multivariate data [6], [8]. PCA was applied to identify latent low-dimensional structure.

The covariance matrix satisfies:

$$\Sigma v_i = \lambda_i v_i$$

where:

- v_i are eigenvectors,
- λ_i are eigenvalues.

A. Explained Variance

As shown in Fig. 8, only a few principal components explain most variance.

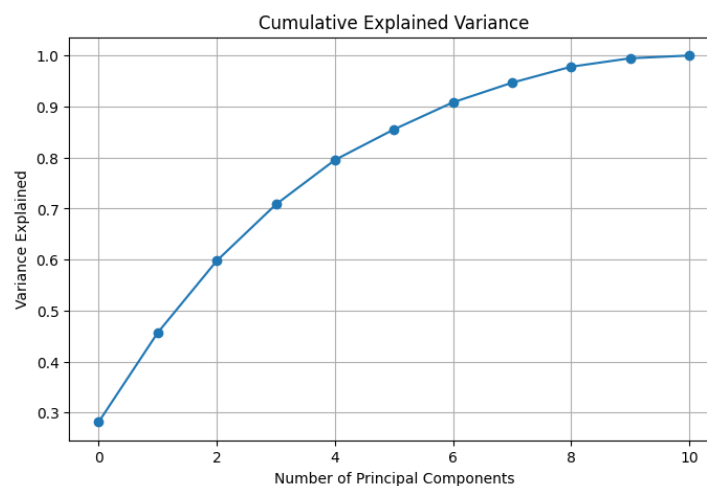


Fig. 8: Explained variance ratio of principal components.

B. PCA Projection

The PCA projection is shown in Fig. 9.

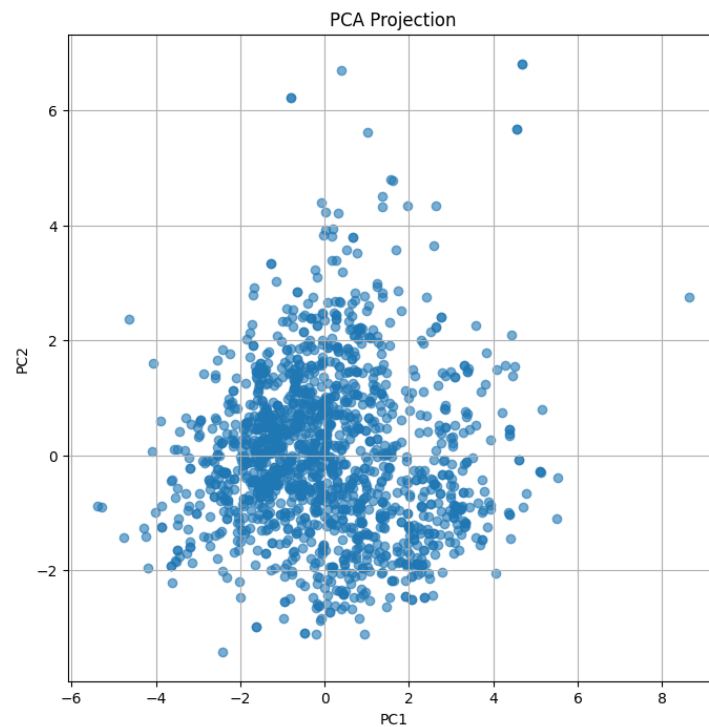


Fig. 9: Projection of wine samples onto the first two principal components [5], [6].

The PCA results reveal latent low-dimensional structure.

XIII. LINEAR REGRESSION

Wine quality was modeled using linear regression. Linear regression models relationships between dependent and explanatory variables using linear combinations of predictors [13], [8]. Regression coefficients provide interpretable estimates of how individual variables influence the target variable [8].

The model is defined as:

$$Y = \beta_0 + \sum_i \beta_i X_i + \epsilon$$

where:

- Y is wine quality,
- X_i are physicochemical variables.

The model achieved moderate predictive performance, indicating partially linear relationships.

XIV. OUTLIER ANALYSIS

Outliers were detected using the interquartile range method.

Significant outliers were observed in:

- sulfur dioxide,
- chlorides,
- residual sugar.

These outliers contribute to deviations from ideal Gaussian assumptions.

XV. DISCUSSION

The analysis demonstrates that the Wine Quality dataset contains rich statistical structure.

Main findings:

- alcohol distribution is non-Gaussian,
- Gamma distribution models skewness better,
- alcohol and density are anticorrelated,
- PCA reveals latent structure,
- linear regression partially predicts quality.

The combination of probabilistic modeling, covariance analysis, and dimensionality reduction provides meaningful insight into the dataset. The observed statistical structure is consistent with previous studies of the Wine Quality dataset, which report meaningful relationships between physicochemical properties and perceived wine quality [1].

XVI. CONCLUSION

This project investigated statistical structure in the Wine Quality dataset using methods introduced in the Foundations of Statistical Modelling course.

The project combined:

- distribution analysis,
- PMFs,
- moment analysis,
- correlation analysis,
- PCA,
- regression modeling.

The results demonstrate that real-world datasets exhibit non-Gaussian behavior, hidden dependencies, and latent structure.

XVII. OUTLOOK

Future work could include:

- nonlinear machine learning models,
- clustering algorithms,
- neural networks,
- larger datasets,
- time-series extensions.

XVIII. REFERENCES

GitHub Project Link: <https://github.com/Subiswas36218/PSM-Wine-Quality-Analysis>

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